

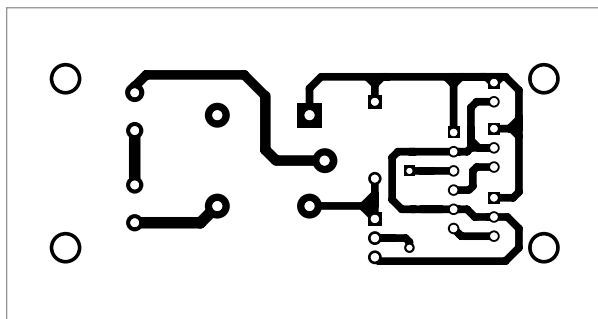


Fig. 6: Actual-size PCB layout for the automated light

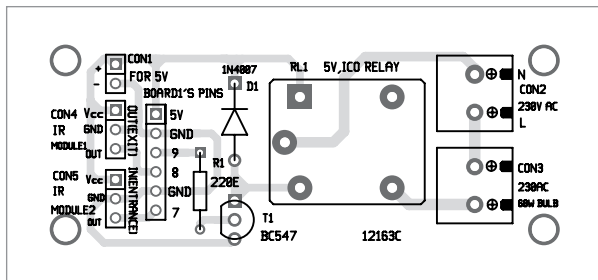


Fig. 7: Components layout of the PCB



Fig. 8: Sensors' installation near the door

person followed by 'out' sensor (Module1), the microcontroller in Arduino assumes that a person is entering the washroom and it provides 5V at pin 9 of Board1. The current from the output pin of Arduino is, however, insufficient to energise the relay. So, npn transistor T1 (BC547) is used for amplification to control the relay and switch on the light.

When 'out' sensor detects a person followed by 'in' sensor, the microcon-

troller assumes that a person is leaving and provides 0V at pin 9, which de-energises the relay. This disconnects the neutral and the light bulb switches off.

If only one of the two sensors is used, the current state will not change, so you need both the sensors. The 'in' sensor should be placed just outside the washroom near the door and the 'out' sensor should be placed inside the washroom near the door. You need to install the sensors such that they can easily detect the person entering or leaving the bathroom.

Software

Arduino Uno is used to compute the logics of the two IR sensors. The circuit is controlled through the software loaded in internal memory of the Arduino. The program (bathroom.ino) is written in Arduino programming language. Arduino IDE 1.8.5 is used to compile and upload the program.

ATmega328P on Arduino Uno comes with a pre-programmed boot-loader that allows you to upload a code to it without using an external hardware like programmer kit. In the IDE, the count (total) is globally initialised to zero.

In the code (bathroom.ino), the 'void setup' contains the initialisation of digital pins 7 and 8 as input, and digital pin 9 as output. Serial begin in void setup is for serial communication. Void show () is a function used for serial communication. Void setup is shown in the Fig. 5.

The main part of the program is void loop, which contains the logic to be executed. First the 'in' sensor is checked for obstruction (person). In case of any obstruction, if condition is satisfied and while loop is executed. At this stage, the 'out' sensor is not yet subjected to obstruction.

When the person obstructs the

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- Module1, Module2 - FC-51 IR sensors
- T1 - BC547 transistor
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 220-ohm

Miscellaneous:

- RL1 - 5V, 1CO relay
- CON1 - 2-pin connector
- CON2, CON2 - 2-pin terminal connector
- CON3, CON4 - 3-pin connector
- Light bulb 60W, 230V AC
- Jumper wires

'out' sensor, the output of the sensor is low, so the program exits the while loop and count is incremented. The vice versa is framed for else if part with the count value decremented. Show () function is used for serial monitor and is unnecessary for practical application. Void loop is shown in Fig. 5.

It is assumed that only one person would be using the washroom at a time, so using if condition, when count (here 'total') is checked for the value '1' is shown in Fig. 5. If total is one, digital pin 9 output will be high only for count one and the relay will be connected. If not, it will be kept low, that is, the relay will not be energised.

Construction and testing

An actual-size PCB layout for the automated light is shown in Fig. 6 and its components layout in Fig. 7. After assembling the circuit on PCB and uploading the source code (bathroom.ino) into Arduino Uno, connect a 5V supply across CON1, 230V AC across CON2, and the light bulb across CON3. Installation of the sensors near the door is shown in Fig. 8. **EFY**



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